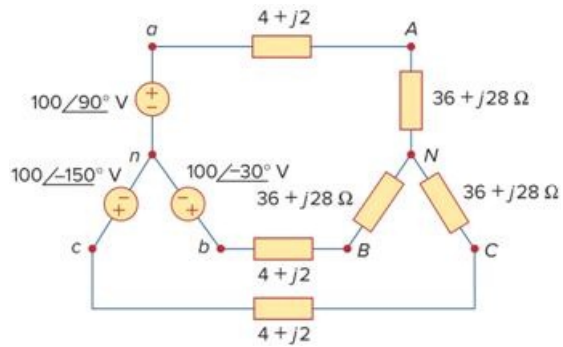


Problem

Find the real power absorbed by the load in Fig. 12.58.

Figure 12.58



Step-by-step solution

Step 1 of 7

Refer to Figure 12.58 from the textbook.

Consider the direction of the loop currents, I_1 and I_2 as shown in Figure 1.

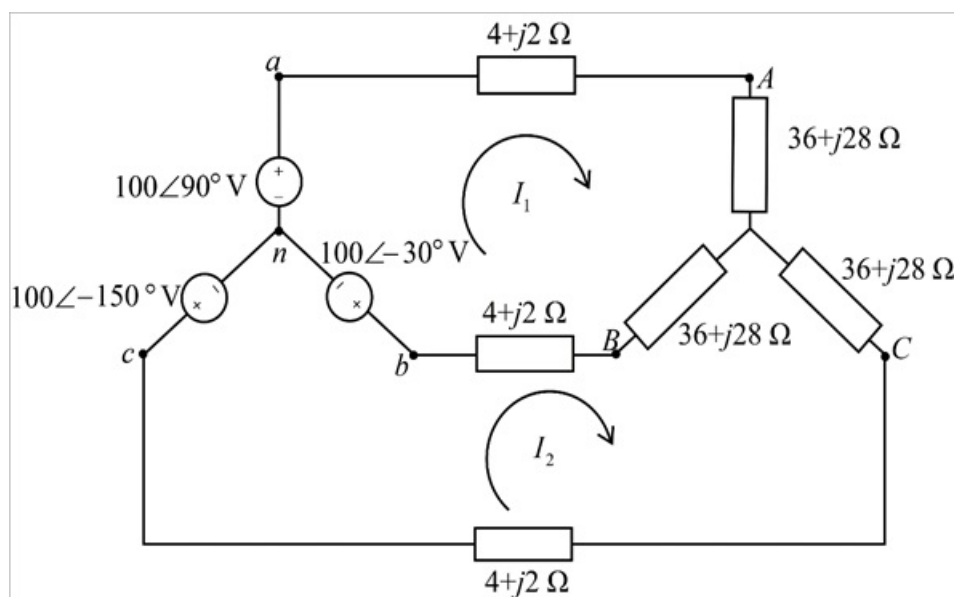


Figure 1

Step 2 of 7

Apply the Kirchhoff's Voltage Law in the loop $aABbn$.

$$\begin{aligned}
 0 &= (4 + j2)I_1 + (36 + j28)I_1 + (4 + j2)(I_1 - I_2) \\
 &\quad + (36 + j28)(I_1 - I_2) - 100\angle 90^\circ + 100\angle -30^\circ \\
 0 &= (2(4 + j2) + 2(36 + j28))I_1 - (4 + j2 + 36 + j28)I_2 \\
 &\quad - 100\angle 90^\circ + 100\angle -30^\circ \\
 0 &= (80 + j60)I_1 - (40 + j30)I_2 - 100\angle 90^\circ + 100\angle -30^\circ
 \end{aligned}$$

Thus,

$$\begin{aligned}
 100\angle 90^\circ - 100\angle -30^\circ &= (80 + j60)I_1 - (40 + j30)I_2 \\
 173.205\angle 120^\circ &= (80 + j60)I_1 - (40 + j30)I_2 \dots (1)
 \end{aligned}$$

Step 3 of 7

Apply the Kirchhoff's Voltage Law in the loop $cnbBC$.

$$\begin{aligned}
 0 &= (4 + j2)I_2 + (36 + j28)I_2 + (4 + j2)(I_2 - I_1) \\
 &\quad + (36 + j28)(I_2 - I_1) - 100\angle -30^\circ + 100\angle -150^\circ \\
 0 &= (2(4 + j2) + 2(36 + j28))I_2 - (4 + j2 + 36 + j28)I_1 \\
 &\quad - 100\angle -30^\circ + 100\angle -150^\circ
 \end{aligned}$$

Thus,

$$\begin{aligned}
 100\angle -30^\circ - 100\angle -150^\circ &= (80 + j60)I_2 - (40 + j30)I_1 \dots (2) \\
 173.205\angle 0^\circ &= (80 + j60)I_2 - (40 + j30)I_1 \\
 I_2 &= \frac{173.205\angle 0^\circ + (40 + j30)I_1}{80 + j60} \\
 I_2 &= 1.386 - j1.039 + 0.5I_1
 \end{aligned}$$

Step 4 of 7

Substitute $1.386 - j1.039 + 0.5I_1$ for I_2 in equation (1).

$$\begin{aligned}
 173.205\angle 120^\circ &= (80 + j60)I_1 - (40 + j30)(1.386 - j1.039 + 0.5I_1) \\
 -86.6 + j149.99 &= (80 + j60)I_1 - (86.61 + j0.02) - (20 - j15)I_1 \\
 -86.6 + j149.99 + 86.61 + j0.02 &= (60 + j45)I_1 \\
 j150 &= (60 + j45)I_1
 \end{aligned}$$

Thus,

$$\begin{aligned}
 I_1 &= \frac{j150}{60 + j45} \\
 &= 2\angle 53.13^\circ \text{ A}
 \end{aligned}$$

Step 5 of 7

Calculate the magnitude of the current, $|I_1|$.

$$|I_1| = 2 \text{ A}$$

Step 6 of 7

Calculate the real power (p_1) delivered to single-phase load of $(36 + j28) \Omega$.

$$p_1 = |I_1|^2 R$$

Substitute 2 A for $|I_1|$ and 36Ω for Z .

$$\begin{aligned} p_1 &= (2)^2 (36) \\ &= (4)(36) \\ &= 144 \text{ W} \end{aligned}$$

Step 7 of 7

Calculate the total power delivered to the load, p .

$$p = 3p_1$$

Substitute 144 W for p_1 .

$$\begin{aligned} p &= (3)(144) \\ &= 432 \text{ W} \end{aligned}$$

Therefore, the real power delivered to the load, p is 432 W.